

LETHBRIDGE, AB, Canada

WMO: 712670

Lat: 49.626N

Lon: 112.794W

Elev: 3048

StdP: 13.15

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-14.0	-28.7	1.2	-20.0	-21.8	1.8	-13.7	41.3	40.2	37.9	40.2	6.7	230	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	89.0	62.0	85.3	61.2	81.8	60.4	65.9	80.8	63.9	79.3	62.1	77.7	11.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	88.8	72.4	58.4	81.7	69.9	56.4	75.9	67.7	32.4	80.8	30.8	79.2	29.4	78.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.3	30.8	27.2	DB	-30.6	95.0	6.6	3.8	-35.3	97.7	-39.1	100.0	-42.8	102.1	-47.6	104.8	(4)
(5)			WB	-30.8	70.3	6.5	2.9	-35.5	72.4	-39.3	74.1	-42.9	75.8	-47.6	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	21.0	22.6	31.1	41.5	51.1	58.4	65.0	63.8	55.5	43.3	30.8	21.8		
	DBStd	19.61	17.03	15.94	14.55	9.10	7.90	5.83	5.51	6.15	8.28	9.90	13.95	15.75		
	HDD50	4365	899	769	590	276	82	6	1	1	41	245	580	875		
	HDD65	8456	1363	1188	1052	706	435	211	68	97	298	672	1026	1340		
	CDD50	1538	1	0	3	20	115	257	466	428	205	39	4	0		
	CDD65	154	0	0	0	0	3	12	68	59	12	0	0	0		
	CDH74	3303	0	0	1	10	134	266	1252	1216	411	13	0	0		
	CDH80	1129	0	0	0	0	24	58	449	468	130	0	0	0		
	WSAvg	11.5	12.9	11.8	12.0	12.1	11.4	10.8	9.6	9.1	10.4	12.3	12.8	13.1		
	PrecAvg	16.7	0.9	0.6	1.1	1.6	2.1	2.6	1.8	1.7	1.8	0.7	0.7	0.9		
(15) Precipitation	PrecMax	30.2	2.3	1.5	3.1	6.7	5.3	6.7	4.8	5.6	5.5	1.8	2.0	2.3		
	PrecMin	9.6	0.2	0.1	0.2	0.0	0.4	0.1	0.3	0.2	0.2	0.0	0.0	0.2		
	PrecStd	4.4	0.5	0.3	0.6	1.3	1.2	1.7	1.3	1.3	1.4	0.5	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	56.9	64.8	75.2	83.1	86.5	93.7	94.1	89.6	75.9	64.4	54.0		
		MCWB	43.4	42.7	46.6	52.8	57.9	62.6	63.7	62.4	59.4	53.6	47.6	41.4		
	2%	DB	49.4	51.1	59.4	68.7	77.8	80.6	89.0	89.2	83.8	69.5	57.9	49.0		
		MCWB	39.6	39.8	44.6	49.5	56.2	60.7	63.6	61.0	57.8	51.5	44.9	38.8		
	5%	DB	45.6	46.7	55.0	63.4	72.8	76.6	85.1	85.3	78.8	64.2	53.2	45.3		
		MCWB	37.1	37.5	42.3	46.9	54.0	59.0	63.0	60.7	56.3	49.3	42.4	36.7		
	10%	DB	42.3	42.9	50.3	58.5	68.2	72.9	81.3	81.3	73.3	59.1	48.7	41.8		
		MCWB	35.4	34.9	39.6	44.4	51.6	57.0	62.4	59.7	54.7	46.3	39.5	34.4		
	0.4%	WB	44.1	43.9	48.3	53.7	59.9	66.4	69.8	66.3	61.8	55.8	49.9	43.1		
		MCDB	53.7	55.3	62.0	71.7	78.3	79.8	84.8	83.1	82.3	71.6	62.4	51.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.0	40.6	45.4	50.5	57.2	63.0	67.1	64.1	59.2	52.5	45.6	39.5		
		MCDB	48.4	50.1	57.9	67.1	74.7	76.0	81.3	81.5	78.8	67.4	56.7	47.8		
	5%	WB	37.7	37.8	42.8	47.7	54.9	60.5	65.2	62.5	57.3	49.8	42.7	37.0		
		MCDB	44.9	46.3	54.2	62.0	70.7	73.5	79.8	79.8	75.6	63.0	52.2	44.6		
	10%	WB	35.6	35.2	39.9	44.9	52.7	58.3	63.4	60.9	55.2	47.1	39.9	34.6		
		MCDB	41.9	42.6	49.6	57.2	66.3	70.4	78.3	77.7	72.1	58.3	48.3	41.6		
	5% DB	MDBR	19.8	20.8	21.7	23.6	24.2	23.3	28.1	29.1	27.0	23.3	20.3	19.4		
		MCDBR	20.6	24.0	28.5	33.4	33.0	30.9	35.7	37.8	37.7	30.9	25.1	20.3		
		MCWBR	13.9	15.4	16.9	18.0	16.0	14.8	15.6	15.5	16.7	16.5	15.6	13.7		
		MCWBR	20.5	24.2	28.2	31.2	30.2	27.9	31.0	32.5	34.4	29.5	24.6	20.5		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.259	0.284	0.320	0.333	0.330	0.333	0.350	0.312	0.280	0.257	0.234		
		taud	2.454	2.486	2.481	2.388	2.406	2.460	2.465	2.393	2.471	2.540	2.503	2.456		
	Ebn at Noon		264	289	298	294	293	294	292	281	283	277	259	256		
		Edh at Noon	19	23	28	34	35	34	33	34	28	22	18	16		
(44) All-Sky Solar Radiation	RadAvg		393	711	1110	1536	1830	2038	2241	1781	1265	788	441	312		
	RadStd		33	55	93	103	154	148	116	109	105	80	35	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page